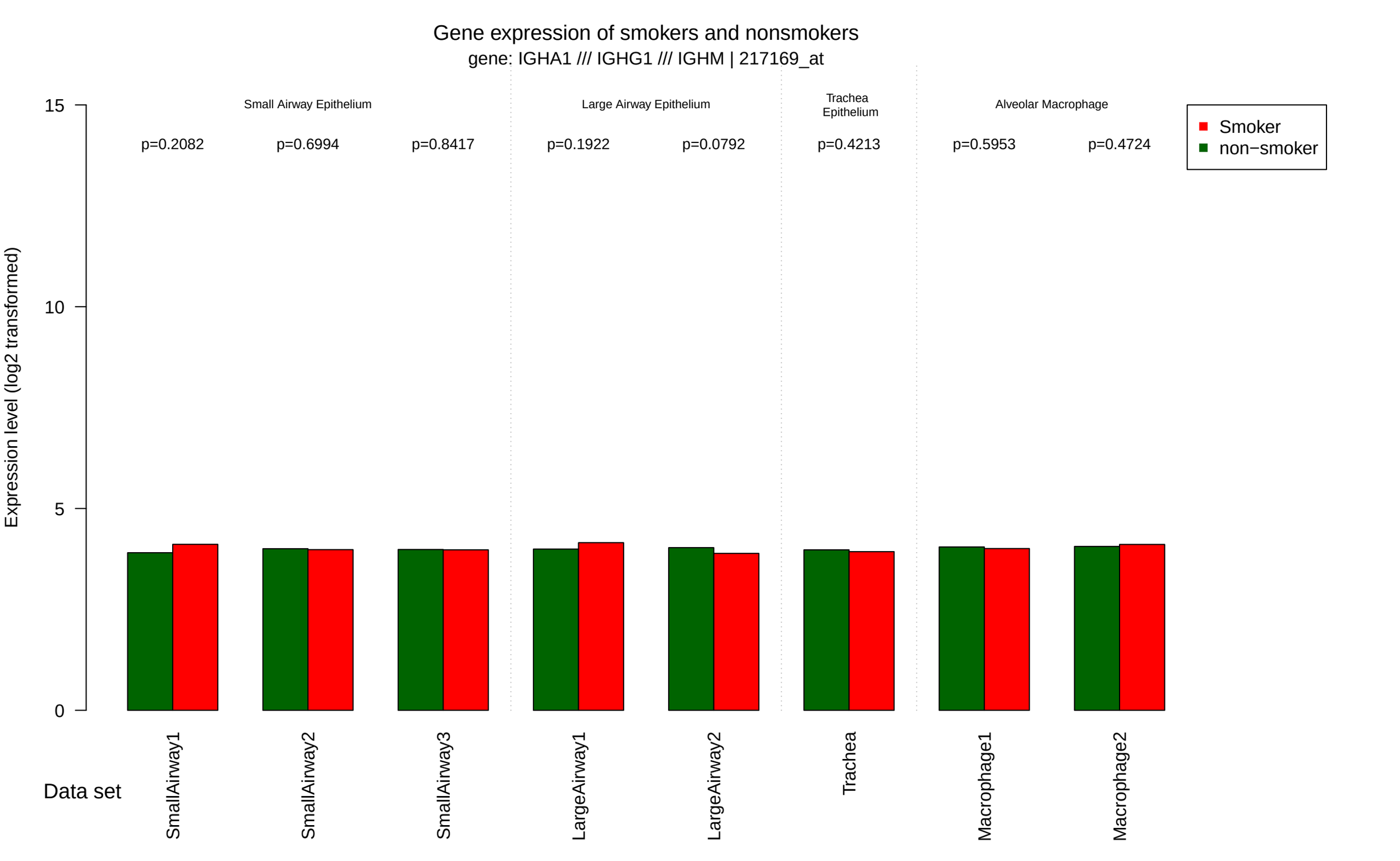
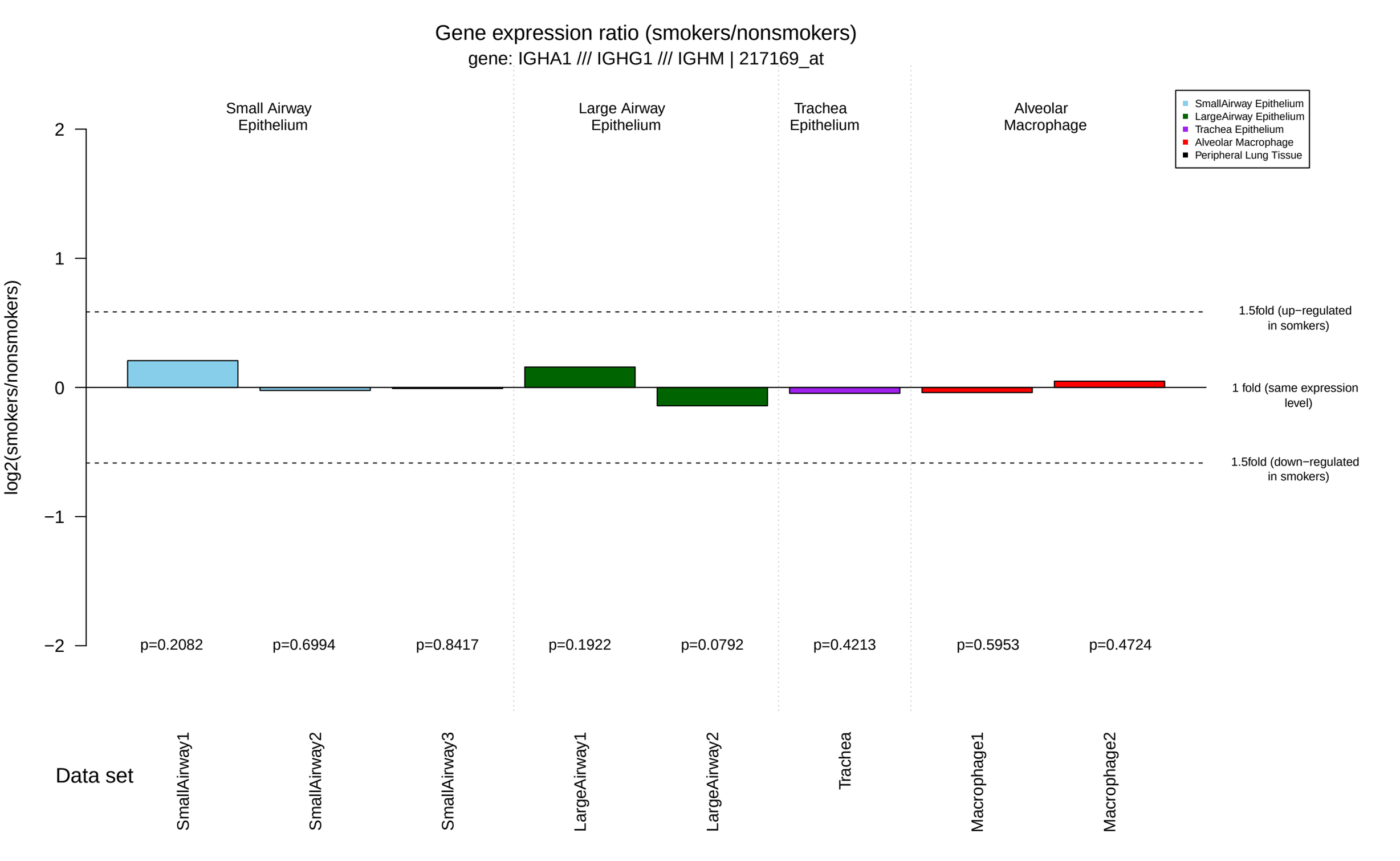
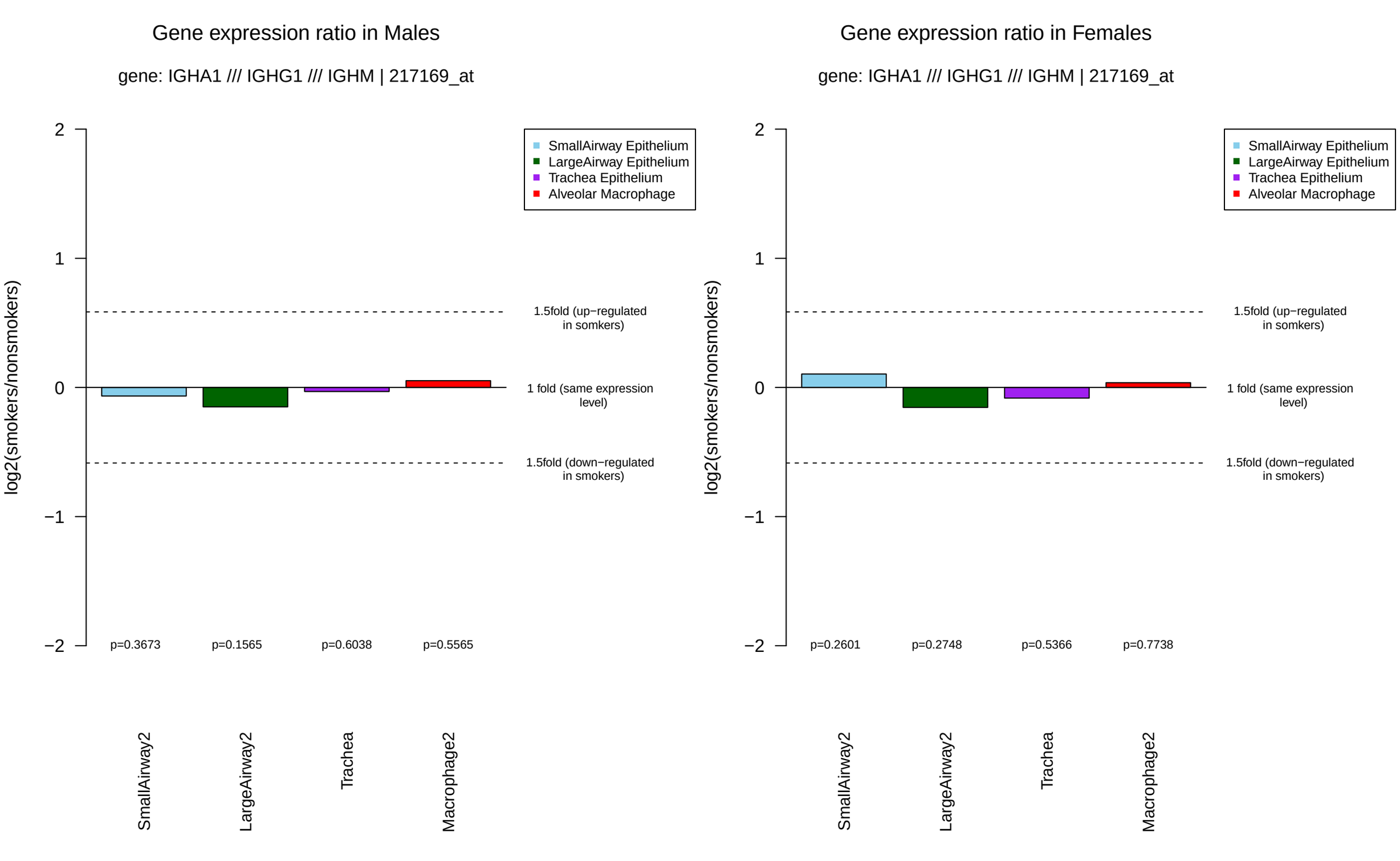
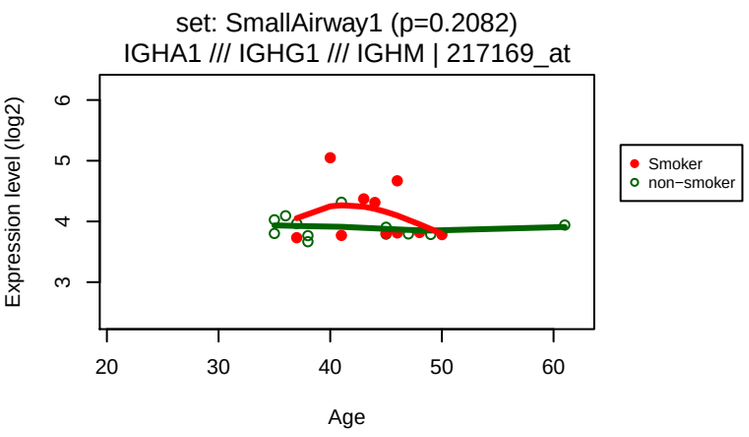
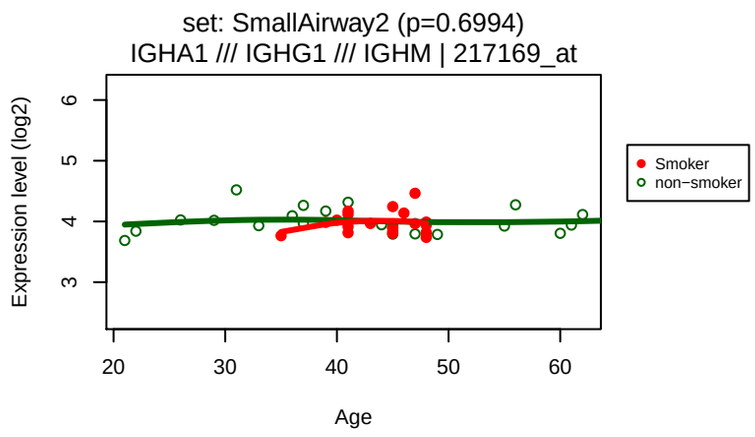
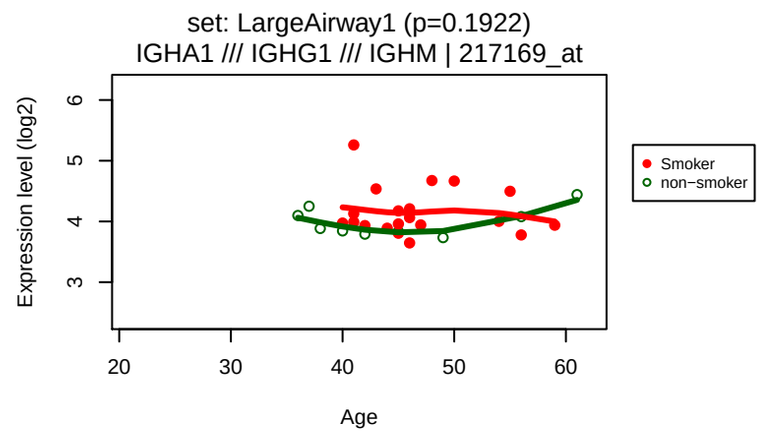

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHG1 /// IGHM | 217169_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2082
p=0.6994
p=0.8417
p=0.1922
p=0.0792
p=0.4213
p=0.5953
p=0.4724
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

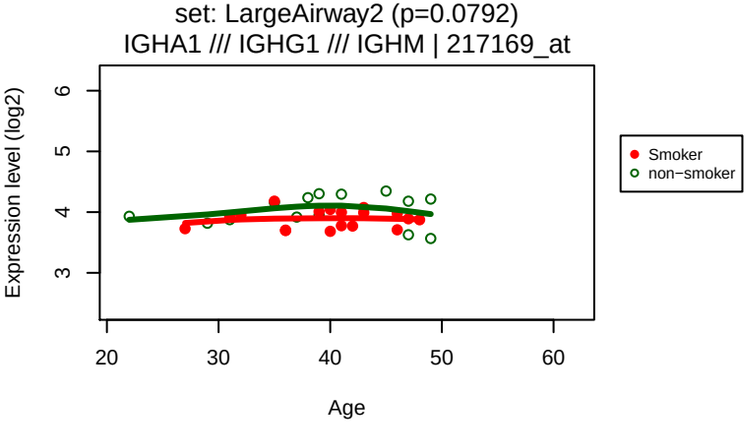

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHG1 /// IGHM | 217169_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2082
p=0.6994
p=0.8417
p=0.1922
p=0.0792
p=0.4213
p=0.5953
p=0.4724
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

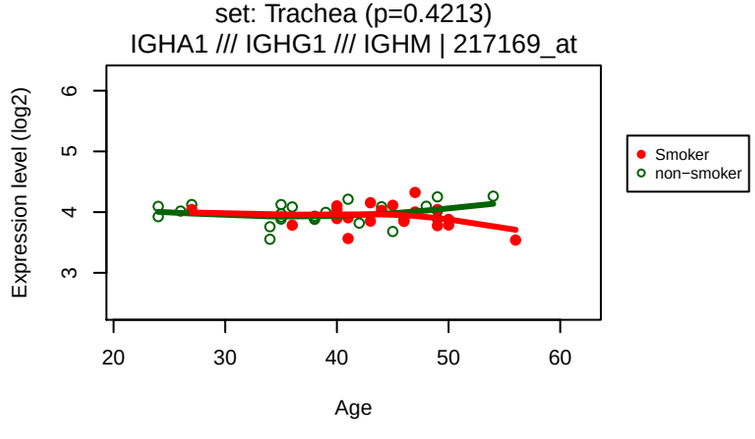

Gene expression ratio in Males
gene: IGHA1 /// IGHG1 /// IGHM | 217169_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3673
p=0.1565
p=0.6038
p=0.5565
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHA1 /// IGHG1 /// IGHM | 217169_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2601
p=0.2748
p=0.5366
p=0.7738
SmallAirway2
LargeAirway2
Trachea
Macrophage2

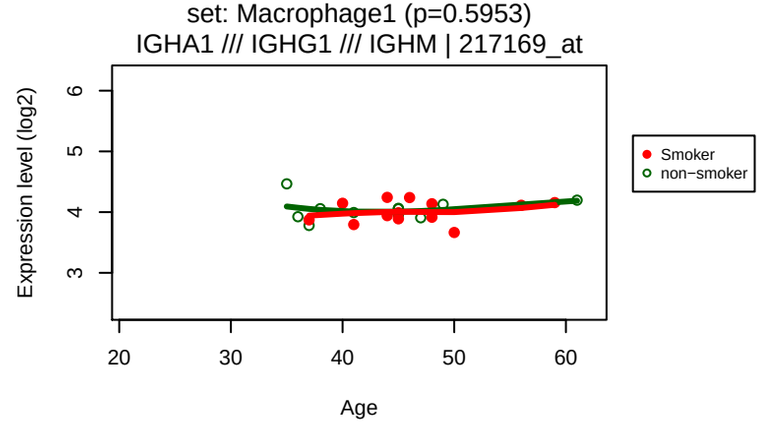

set: SmallAirway1 (p=0.2082)
IGHA1 /// IGHG1 /// IGHM | 217169_at
Expression level (log2)
Age
Smoker
non−smoker

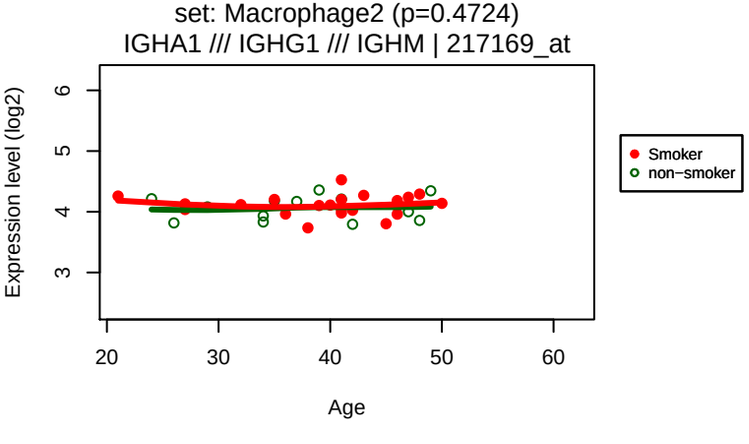

set: SmallAirway2 (p=0.6994)
IGHA1 /// IGHG1 /// IGHM | 217169_at
Expression level (log2)
Age
Smoker
non−smoker

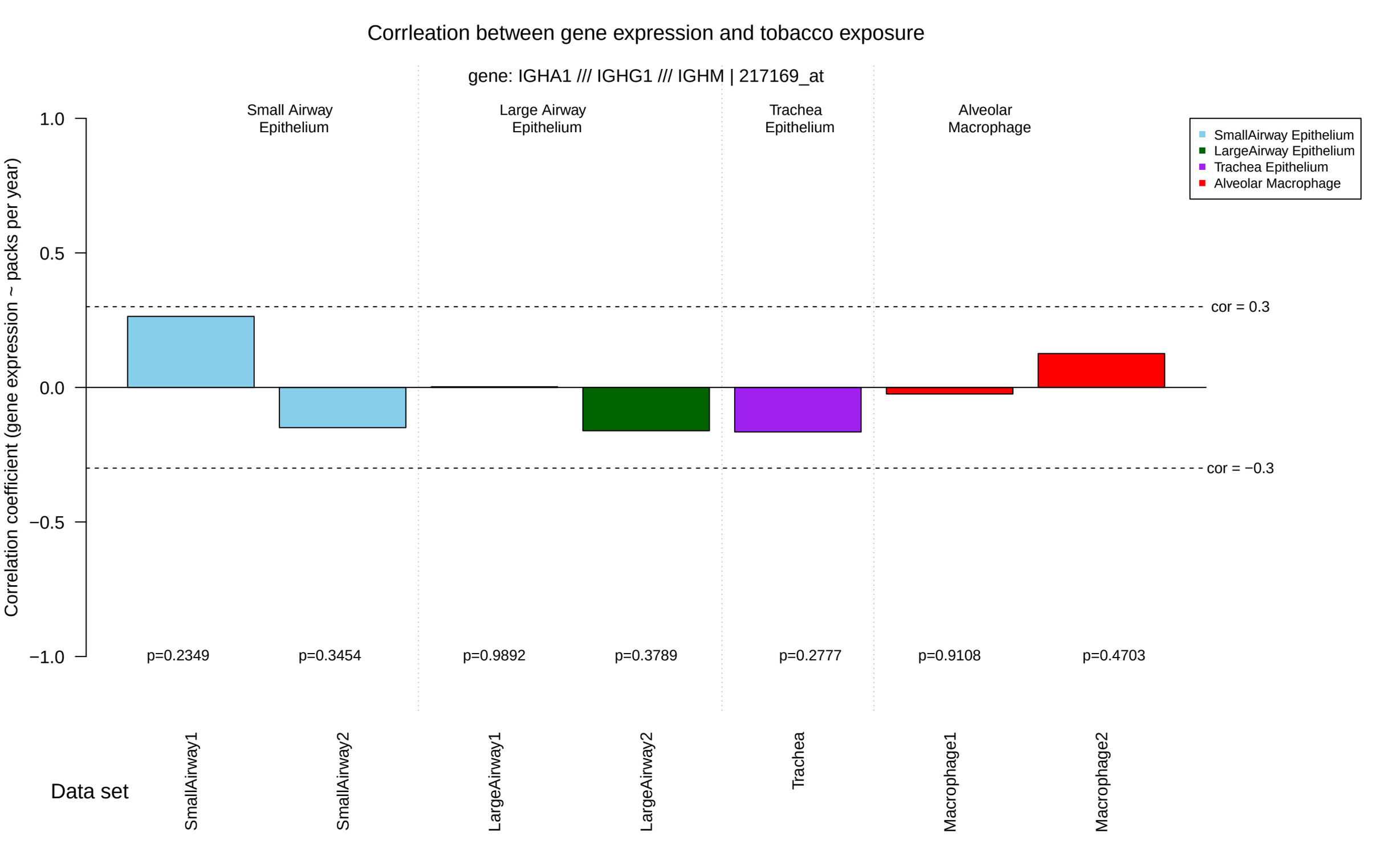

set: LargeAirway1 (p=0.1922)
IGHA1 /// IGHG1 /// IGHM | 217169_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0792)
IGHA1 /// IGHG1 /// IGHM | 217169_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4213)
IGHA1 /// IGHG1 /// IGHM | 217169_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5953)
IGHA1 /// IGHG1 /// IGHM | 217169_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4724)
IGHA1 /// IGHG1 /// IGHM | 217169_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHG1 /// IGHM | 217169_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2349
p=0.3454
p=0.9892
p=0.3789
p=0.2777
p=0.9108
p=0.4703
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2